# QCalc design documents (v1.4.0)

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This is the overview of the design documents of **QCalc 1.4.0** (versionCode 6, package name `io.github.hiroshitsu.qcalc`): its requirements, design, use, maintenance and tests, written to match the current source code. Every document is in English and in Japanese, with the same content: the English tabs are this overview and tabs 1 to 6 in English, and the Japanese tabs are 概要 and tabs 1 to 6 in Japanese; section 11 lists them all. References such as “detailed design 3.6” point to the section where a topic is described in detail.

## 1. What QCalc is

QCalc is a multi-function calculator for Android. It runs **libqalculate 5.13.2**, the calculation engine of the open-source desktop calculator Qalculate!, without modifying it, and adds a user interface for phones. The user interface is in English (the default) and Japanese.

- **Why**: Qalculate! has versions for PCs (GTK, Qt and the command-line `qalc`) but no official Android version. The goal is to make all of its calculation features usable on an arm64 Android phone and to publish the app on Google Play. The reference device is a SHARP AQUOS sense8 (Android 16, 1080 × 2432).
- **What users do**: type expressions such as `60 mph to km/h`, `100 USD to JPY`, `solve(x^2 - 5x + 6 = 0)` or `integrate(x^2 sin(x), 0, pi)`. The syntax and the results are the same as those of `qalc`.
- **What QCalc adds to libqalculate**: only ten traditional Japanese units (坪, 反, 寸, 尺, 間, 里, 合, 升, 匁, 貫) and the matrix trace `trace()`, both added to the running engine at start-up.
- **What 1.4.0 changed**: a review of the whole app made it easier to use and fixed the problems that were found. Suggestions appear under the input field: after → (to), the units to convert to, and while typing with the phone keyboard, the names of functions, units and constants (Japanese names are found too). A new setting, Light or dark, keeps the light or the dark colors whatever the phone's setting. The menu of a history line shows when it was calculated, and a new result is added to the history without building the whole list again. Three problems were fixed: “to mixed” gave results with 寸 or 貫, unlike the official qalc; an expression ending with “to” showed a meaningless preview; and in landscape a long expression pushed the keys off the screen. The table “Changes in 1.4.0” below lists the changes.
- **What 1.3.0 changed**: the screens are in English and Japanese, with English as the default (“Language / 言語” in the menu or in Settings switches them). QCalc Pro, a one-time purchase in Google Play, unlocks precision above 1,000 digits, 3D graphs,

seven color themes and exporting the history as a CSV file; everything else stays free. Builds without the Billing Library, such as the 1.3.0 App Bundle made by the shell scripts, lock nothing, and users from before Pro keep the Pro features for free. The live preview now shows all the digits like the history (a very long result shows its beginning and its end), and digit grouping puts commas only in the integer part. The table “Changes in 1.3.0” below lists the changes.

- **What 1.2.0 changed**: the precision limit was raised from 1,000 to 200,000 digits, the most that works well on Android (the clipboard takes about 1 MB at a time). Long results are shown, copied and kept in full: the history keeps the newest results whole up to 1,000,000 characters in total. ans is given back with all its digits after a restart (1.1.0 rounded it to about 38 digits), and long numbers in it are read directly with GMP. A result too long to copy gives a message instead of a crash.
- **What 1.1.0 changed**: four keypads switched with tabs (123, functions, calculus and matrices), matrix results shown as tables, built-in examples (12 topics, 69 expressions), undo for a cleared expression or history, three text sizes, Japanese names and search for functions, units and constants, and a workaround for a libqalculate 5.13.2 problem that hid later warnings (section 5).

| In scope | Out of scope |
|---|---|
| Everything libqalculate calculates: arithmetic and functions, units, currencies, equations ( `solve` ), derivatives and integrals, matrices, number bases, dates, uncertainty propagation, complex numbers, data sets, up to 200,000 digits (above 1,000 with QCalc Pro) | Devices other than arm64 (32-bit ARM, x86) |
| Input with the on-screen keys and the phone keyboard, live preview, history, `ans` , cancelling a calculation | gnuplot plotting and the file and command functions such as `command()` , `load()` and `export()` (QCalc draws its own graphs) |
| Lists of functions, variables and units with search, 2D and 3D graphs, user-defined variables and functions, exchange-rate updates, settings, examples, color themes, CSV export of the history | RPN mode, user interfaces in languages other than English and Japanese, sync between devices |
| Distribution on Google Play, QCalc Pro as a one-time purchase, license texts in the app, published source code | Eigenvalues as a single function, differential equations above first order, values of infinite sums (libqalculate cannot do these) |

**Changes in 1.4.0.** The page (app.js, style.css, index.html), the Java code (MainActivity.java and Pro.java) and the definitions of two Japanese units in the bridge program qalc-engine.cc (two lines) changed. libqalculate itself and the protocol between the app and the engine (the commands and their form) did not.

| Change | What | Details in |
|---|---|---|
| Suggestions | A row of suggestions under the input field. After “to”: units of the same kind as the unit of the expression (without a unit, ways to write a number). While typing with the phone keyboard: names of functions, variables and units. Made by the page alone, with no new engine command | Requirements definition 1.5 (F-35); detailed design 3.24; user manual 4.3 and 4.6 |
| Light or dark | The setting appearance (same as the phone, light, dark). JsApi gained setAppearance (15 methods), and the colors of the system bars follow | Detailed design 3.2 and 3.25; user manual 4.9 |
| History | The menu of a line begins with the date and time of the calculation. On confirm, only the new line is added to the list on the screen | Detailed design 3.9 |
| “to mixed” fixed | The relations of 寸 and 貫 are written in another way (with the same values), so that libqalculate no longer picks them as units to mix. “1.5 m to mixed” had given “16.5 寸 + 1 m” | Detailed design 3.19 |
| “to” at the end of an expression | The preview and confirm calculate the expression without the trailing “to” (“255 to” had given “255 B·t”) | Detailed design 3.8 |
| Landscape layout | Each half keeps half of the screen (a long expression had pushed the keys off the screen) | Basic design 2.5 |

| Change | What | Details in |
|---|---|---|
| Others | The page is told about the state of QCalc Pro only when it changes. The zoom buttons of the graphs are labeled + and −. Results wrap between words, and only a number longer than a line is cut at the end of the line (3.9) | Detailed design 3.20 |
| Version | versionCode 6 | Build and maintenance manual 5.7 |
| Tests | 27 new page tests (T01 to T27) and 8 expressions with “to mixed” in the comparison. The comparison with qalc (317 expressions), the engine tests, the examples of both screens, the 31 Java tests, the tests on the Japanese screen, the long-result and history page tests, the page tests of 1.3.0 and the 22 package checks were run again. 700 items in all, all passed | Test specification and results |

**Changes in 1.3.0.** The page ( `app.js` , `style.css` , `index.html` ), the Java code ( `MainActivity.java` , the new `Pro.java` , and `PlayBilling.java` , which only builds with billing contain) and a part of the bridge program `qalc-engine.cc` ( `tail_chars` , which returns the end of a long result) changed. libqalculate itself did not.

| Change | What | Details in |
|---|---|---|
| Language of the screen | English and Japanese. English is the default, and a language the user chooses is saved in `qalc.lang` . The engine and the Java code report in English; on the Japanese screen the page translates their messages (57 kinds) | Requirements definition 1.5 (F-30); detailed design 3.14 and 3.21 |

| Change | What | Details in |
|---|---|---|
| QCalc Pro | A one-time product in Google Play ( `qcalc_pro` ). Precision above 1,000 digits, 3D graphs, color themes and exporting the history are Pro features. Users of builds without billing get them for free (legacy) | Requirements definition 1.9; detailed design 3.20 |
| Color themes | Seven colors for the keys and highlights, in both the light and the dark theme | Detailed design 3.22 |
| Exporting the history | A CSV file (UTF-8 with a byte order mark), protected so that spreadsheets never run a cell as a formula | Detailed design 3.23 |
| All digits in the preview | While typing, results of up to 450,000 bytes are shown; a longer one keeps its beginning and its end ( `tail_chars` ) | Detailed design 3.5, 3.8 and 3.9 |
| Digit grouping | Commas only in the integer part; the decimals and the space before a unit keep narrow spaces | Detailed design 3.9 |
| Version | versionCode 5 (4 was used by an App Bundle built before English became the default) | Build and maintenance manual 5.7 |
| Tests | 27 new page tests (Q01 to Q27). The comparison with qalc (309 expressions), the engine tests, the examples of both screens, the 31 Java tests, the tests on the Japanese screen and the 22 package checks were run again. 665 items in all, all passed | Test specification and results |

## 2. Requirements in brief

Requirements definition 1.5 lists 35 functional requirements (F-01 to F-35; F-22 to F-28 were added in 1.1.0, F-29, long results, in 1.2.0, F-30 to F-34 in 1.3.0, and F-35, suggestions while typing, in 1.4.0). F-01 to F-33 and F-35 are met in 1.4.0; F-34, buying QCalc Pro, works

only in builds with the Google Play Billing Library (requirements definition 1.9).

- **Calculation** (F-01): the full Qalculate! feature set with the same syntax. libqalculate 5.13.2 is used unmodified, and the typed expression is passed on as it is.
- **Input** (F-02, F-03, F-14, F-22 to F-24): four 5 × 5 keypads plus eight keys that are always shown, the phone keyboard, and automatic conversion of full-width characters. Calculus keys insert templates such as `integrate(, 0, 1)` with the cursor where the expression goes, and an a=b key inserts the “=” of equations.
- **Results** (F-04 to F-08, F-15): a live preview 160 ms after the last key (cut off after 1.5 s), confirming with “=” (limited to 20 s), a history of up to 300 items, `ans`, reuse and copy of results, and cancelling. Inserting a result uses the exact value returned by the engine.
- **Lists and definitions** (F-09, F-11): lists of functions, variables and units from the engine, which on the Japanese screen can also be searched by Japanese names, and user-defined variables and functions saved in the engine's definition files.
- **Graphs** (F-10, F-19): 2D graphs of up to 6 functions, and 3D graphs of up to 4 surfaces on a 25 × 25, 41 × 41 or 61 × 61 grid, computed by the engine and drawn by the page.
- **Data** (F-12, F-18): exchange rates from the same three sources as Qalculate!, updated by hand or automatically after 7 days. The history, the settings, the draft and `ans` are written to files at once.
- **Usability** (F-13, F-16, F-17, F-25 to F-27, F-31, F-35): 20 settings, light and dark colors (following the phone or chosen in the settings) and seven color themes, a landscape layout, suggestions for conversion targets and names, built-in examples to tap and try, undo, and three text sizes (1, 1.18 and 1.38 times).
- **Distribution** (F-20, F-21): a privacy policy on GitHub Pages, the full license texts of the app and its seven bundled components, and an App Bundle with Play App Signing.
- **Additions** (F-28): the Japanese units and `trace()`, added by the bridge program at start-up and never written to the user's definition files.
- **Long results** (F-29, F-33): results of up to 200,000 digits are shown, copied and kept with all their digits in the history and in ans and, since 1.3.0, also in the live preview. A result over 450,000 bytes shows its beginning and its last 20,000 bytes.
- **Language** (F-30): English by default and Japanese by choice (“Language / 言語” in the menu or in Settings). The page holds both languages; the Java code and the engine report in English, and on the Japanese screen the page translates their common messages (57 kinds).
- **Export** (F-32): the history as a CSV file (UTF-8 with a byte order mark), saved where the user chooses with Android's file picker, without a storage permission.

- **QCalc Pro** (F-34): a one-time purchase in Google Play that unlocks precision above 1,000 digits, 3D graphs, color themes and the CSV export (requirements definition 1.9). Builds without the Billing Library lock nothing, and users from before Pro keep the Pro features for free.

The non-functional requirements (requirements definition 1.6) come down to one idea: whatever is typed, the app must never freeze, and the value shown, the value saved as `ans` and the value inserted from the history must always be the same.

- **Performance**: the preview is cut off after 1.5 s, a confirmed calculation after 20 s, a 2D graph after 12 s and a 3D graph after 15 s. Every request carries its own time limit, and the engine starts together with the page.
- **Reliability**: the app never freezes and recovers from engine crashes, thanks to three layers of time limits (section 6). A request that crashed the engine is never sent again; a request is resent only when it could not be written.
- **Accuracy**: each expression is evaluated once, the results match `qalc` (317 expressions compared), and warnings are never lost, because message stops are closed before every command.
- **Data safety**: files are written to a temporary file, synced with fsync and then renamed. Unreadable definition files are moved aside, never overwritten.
- **Security and privacy**: the WebView shows only the bundled page, `command()` is disabled at build time, and the only network traffic is downloading exchange rates (and, in builds with billing, talking to Google Play). No personal data is collected.
- **Compatibility**: Android 7.0 to 16 with targetSdk 36 (edge-to-edge display on Android 15+, back handling on Android 13+), and a static arm64 executable aligned for 16 KB pages.
- **Maintainability and license**: libqalculate is unmodified, all builds are reproducible with four shell scripts, the app is GPL v2 or later (with a permission to link the Google Play Billing Library), and the sources of every bundled library are published.

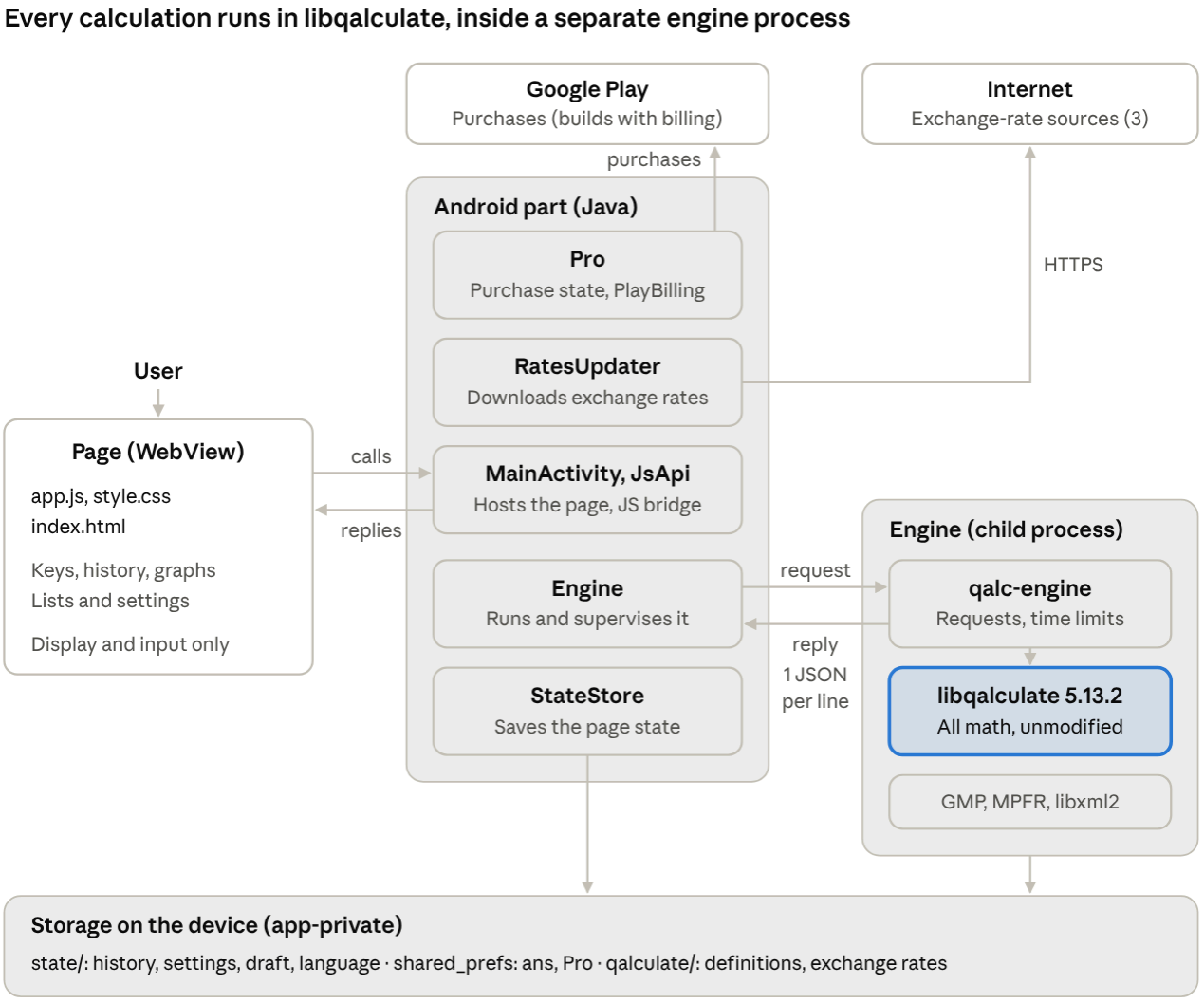
Main constraints (requirements definition 1.7):

1. The source of libqalculate is not changed. Everything specific to QCalc lives in a small bridge program that calls libqalculate's public API.
2. The engine is built with zig as a cross compiler, without the Android NDK. The APK and the App Bundle are assembled with Ubuntu's Android tools (aapt2, dx, zipalign, apksigner, jarsigner) and android.jar from the Android SDK. The build with billing for QCalc Pro is made with Gradle instead (qcalc.billing=true), because the Billing Library comes from Google's Maven repository, which the build environment cannot reach.

3. The engine is a Linux executable stored as `lib/arm64-v8a/libqalcengine.so`. Android 10 and later do not let an app run files from its data directory, so the engine is started from the native library directory, where Android extracts it at install time.
4. targetSdk is 36, as Google Play requires for new apps, so the app supports edge-to-edge display (Android 15+) and the new back handling (Android 16).
5. Google Play signs the app (Play App Signing). The developer keeps only an upload key, which is never published.
6. The name and the icon differ from those of Qalculate!, and the store listing says that QCalc is not an official Qalculate! app.
7. libqalculate has no Japanese translation, so the engine's messages and list names are in English. On the Japanese screen, the page translates the 57 most common messages of the engine and the Java code and the category names, and shows Japanese names beside functions, units and constants.
8. English is the default at every start until a language is chosen, also on phones set to Japanese and for users updating from 1.2.0 (requirements definition 1.7).

Limits of libqalculate itself are explained to users instead of being worked around: `limit()` always warns that its result may be wrong, `dsolve()` solves only first-order equations, infinite sums are not evaluated, there is no eigenvalue function, and one-letter names such as `a` or `m` may be read as units.

# 3. Architecture




Architecture · the page, Java, the engine process and storage

The user only touches the page in the WebView. Every calculation runs in libqalculate inside a separate engine process. The Java layer connects the two, supervises the engine, saves data, downloads exchange rates and keeps the state of QCalc Pro (basic design 2.1–2.3).

The parts meet only through three fixed interfaces, and each of them passes plain strings (JSON). This is what makes it possible to test the page, the Java code and the engine separately (section 9).

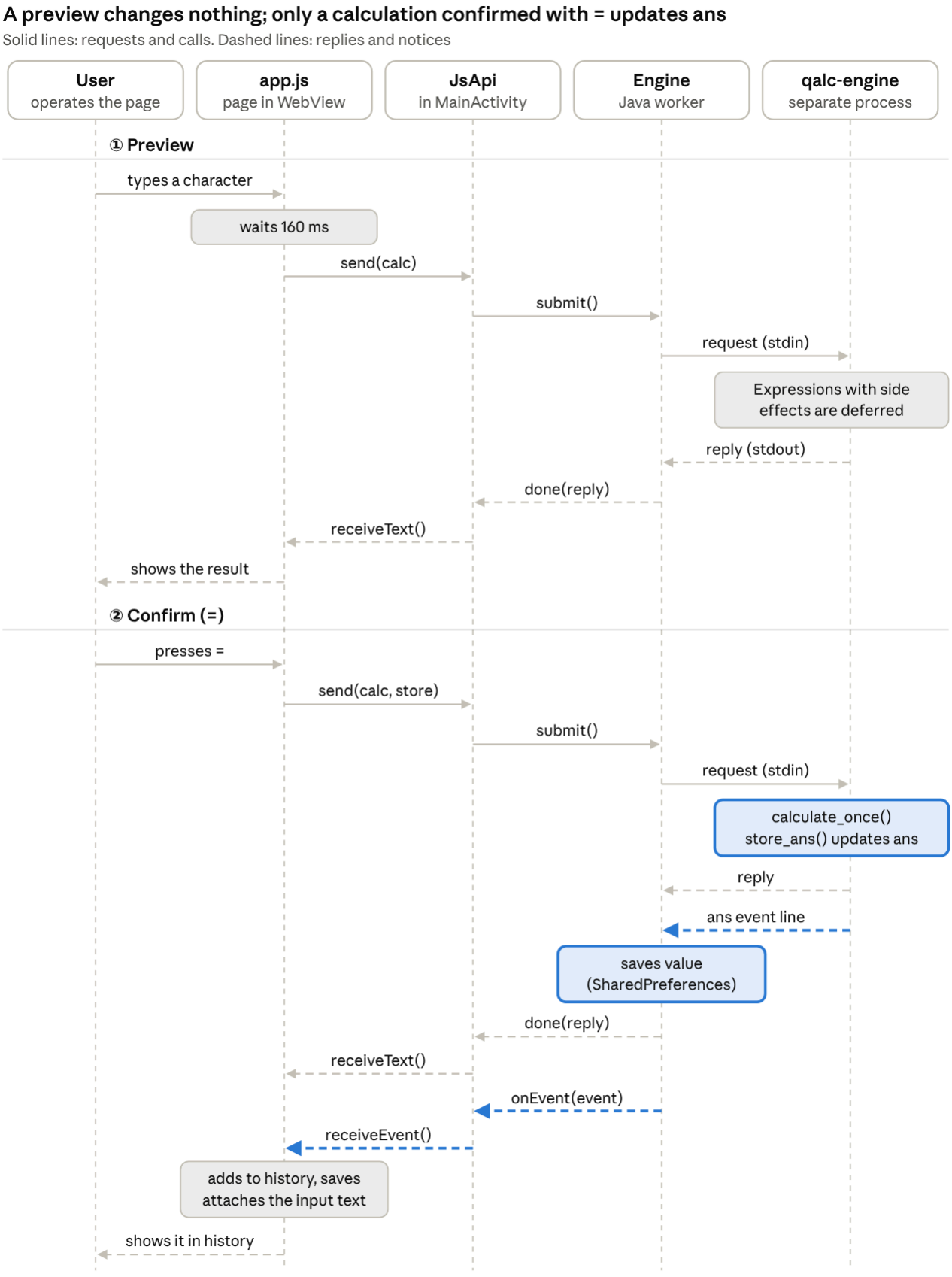
| Interface | Direction | Contents |
|---|---|---|
| `QalcNative` | Page → Java | Fifteen methods that `JsApi` exposes to the WebView: `send`, `cancel`, `updateRates`, `loadState`, `saveState`, `copy`, `appInfo`, `openUrl`, and since 1.3.0 `proInfo`, `buyPro`, `restorePro`, `refreshPro`, `setTestPro` and `saveFile`, and since 1.4.0 setAppearance |
| `QalcBridge` | Java → page | Callbacks provided by `app.js`: `receiveText` for replies and `receiveEvent` for events (`ans`, and since 1.3.0 `pro` when the state of QCalc Pro changes and `saved` with the result of saving a file), called with `evaluateJavascript` |
| JSON lines | `Engine` ⇄ `qalc-engine` | One request per line on the engine's standard input; one reply or event per line on its standard output |

| File | Lines | Role |
|---|---|---|
| `app/assets/index.html` | 52 | Page layout: top bar, history, input field, keypad and menu containers |
| `app/assets/style.css` | 776 | Page look: light and dark colors, seven color themes, keys and tabs, matrix tables, text sizes, panels, graphs, landscape layout |
| `app/assets/app.js` | 2,918 | Page logic: English and Japanese texts, keys and templates, input, suggestions, preview, confirm, history, matrix tables, undo, examples, lists and search, 2D and 3D graphs, definitions, rates, settings, the QCalc Pro panel, CSV export, licenses |
| `MainActivity.java` | 622 | `MainActivity` and `JsApi`: the WebView and its settings, the JavaScript bridge, window insets (Android 15+), back handling, lifecycle, setting up QCalc Pro, saving files |
| `Engine.java` | 357 | Starts, supervises and restarts the engine; matches replies to requests; saves and restores `ans` |
| `RatesUpdater.java` | 164 | Downloads, checks and replaces the exchange-rate files |

| File | Lines | Role |
|---|---|---|
| `StateStore.java` | 123 | Writes the page state (history, settings, draft) to files at once |
| `Pro.java` | 215 | The state of QCalc Pro (owned, pending, earlier users, price) and the kind of billing (play, debug or none); loads PlayBilling by its class name, so that builds without it still work |
| `PlayBilling.java` | 296 | Purchases with the Google Play Billing Library 8: the connection, the price, the purchase screen, acknowledging. Only in builds with billing (kept in app/src-billing) |
| `bridge/qalc-engine.cc` | 1,647 | The engine's bridge: JSON protocol, calls into libqalculate, the beginning and the end of long results, graph values, `ans`, watchdog, protection of definition files, Japanese units and `trace()`, closing message stops |
| libqalculate, GMP, MPFR, libxml2 | — | The engine's core: official sources (5.13.2, 6.3.0, 4.2.2 and 2.15.4), linked statically |

**Deployment** (basic design 2.2). Google Play receives the App Bundle signed with the upload key, builds an APK for each device (about 3.3 MB for arm64) and signs it with the app signing key that Google keeps. When the app is installed, Android extracts `libqalcengine.so` into the native library directory (`android:extractNativeLibs="true"`). Despite its name, it is an executable: `Engine` starts it with `ProcessBuilder` as a child process in the working directory `files/`, with these environment variables: `HOME`; `QALCULATE_USER_DIR` (`files/qalculate/`, where libqalculate keeps user definitions and exchange rates); `TZ` (the current UTC offset in POSIX form, because a static Linux program cannot read Android's time-zone data); `LANG=C.UTF-8`; and `QALC_CURRENCY` (the currency of the phone's region, JPY in Japan).

# 4. Main flow: preview and confirm




Sequence diagram · live preview and confirmed calculation

The live preview and the confirmed calculation are both `calc` requests (detailed design 3.5). They differ only in four fields: `store` (whether `ans` is updated), `preview` (whether expressions with side effects are skipped), `timeout` and the maximum length of the result.

- **Preview**: 160 ms after the last key, the page sends `calc` with a 1.5 s limit. Expressions that would change something — assignments such as `x = 5`, `save()`, or user functions that save — are not run in a preview. The engine answers `deferred` instead, so typing never defines or saves anything by accident.
- **Confirm (=)**: `calc` with `store:true` and a 20 s limit. The engine evaluates the expression exactly once (`calculate_once`: the same steps as `calculateAndPrint` in `qalc`, but with a single evaluation), so random values or expressions like `a = a + 1` are never evaluated twice, and the value shown is the value stored.
- **ans**: after a confirmed calculation, the engine shifts `ans5 ← … ← ans` as `qalc` does and sends an `ans` event line after the reply. Java saves its `value` in SharedPreferences and gives it back with `set_ans` whenever a new engine process starts, so `ans` survives restarts. The event's `input` field is the exact text that the history inserts when the user taps a result (for example `7/3`, not the rounded `2.333333333`).

## 5. The engine protocol

The app writes one JSON request per line to the engine's standard input, and the engine answers with one JSON line on its standard output, with the same `id` (detailed design 3.3). Requests are handled one at a time.

- Success: `{"id":…, "ok":true, …}`. Failure: `{"id":…, "ok":false, "error":"…"}`.
- `"restart":true` in a reply means that the engine exits after the reply; Java starts a new process for the next request.
- Anything that libqalculate prints by itself is moved to standard error at start-up, so it never mixes with the replies.
- Error texts and messages are in English. Before 1.3.0 the bridge program's own texts were in Japanese; now the page translates them on the Japanese screen (detailed design 3.14 and 3.21).

| Command | Main request fields | Main reply fields |
|---|---|---|
| `init` | — | Engine version, time of the exchange rates, whether `ans` is defined |

| Command | Main request fields | Main reply fields |
|---|---|---|
| `calc` | `expr`, `opts` (settings), `timeout` (100–120,000 ms), `store`, `preview`, `max_chars`, and since 1.3.0 `tail_chars` (how much of the end of a long result to return) | `result`, `parsed`, `approx`, `comparison`, `timeout`, `truncated`, `chars` and `digits` (when cut), `tail` (since 1.3.0), `messages`, `deferred` |
| `set_ans` | `expr` | `set` |
| `plot` | `exprs` (up to 12), `min`, `max`, `steps`, `var`, `timeout` | `series` (x and y values for each expression), `messages` |
| `plot3d` | `exprs` (up to 4), `xmin`, `xmax`, `ymin`, `ymax`, `nx` and `ny` (2–121), `timeout` | `x`, `y`, `series` (z values on the grid, `complete`), `messages` |
| `catalog` | `kind` (functions, variables or units) | `items`: names, titles, categories, descriptions and the text to insert |
| `rates_info`, `rates_reload` | — | Sources and file names; whether new rates were loaded |
| `define_var`, `define_func`, `delete_def`, `user_defs` | A name and a value or an expression | `saved`; the list of user definitions |

```
{"id":19,"cmd":"calc","expr":"7/3","opts":
{"approx":"auto","precision":10},"timeout":20000,"store":true,"max_chars":450
000,"tail_chars":20000}
{"id":19,"ok":true,"result":"2 + 1/3 ≈ 2.333333333","parsed":"7 /
3","approx":false,"comparison":false,"timeout":false,"truncated":false,"messa
ges":[]}
{"event":"ans","id":19,"value":"7/3","input":"7/3"}
```

**A workaround for libqalculate 5.13.2** (detailed design 3.19). After a definite integral with an infinite bound, libqalculate leaves one level of “message stop” (`beginTemporaryStopMessages`) open, and from then on no warning or error is shown; the official `qalc` behaves the same way. The bridge program closes any open message stop before every command, so QCalc keeps showing warnings such as division by zero. libqalculate itself is not changed.

# 6. Time limits and recovery

Time limits work in three layers (detailed design 3.13). The innermost one normally ends a request; the outer ones are safety nets.

| Request | Engine limit ( `timeout` ) | Engine watchdog (+3 s) | Java wait (timeout × expressions + 10 s) | Page's last safety net |
|---|---|---|---|---|
| Preview | 1.5 s | 4.5 s | 11.5 s | 120 s |
| Confirm (=) | 20 s | 23 s | 30 s | 120 s |
| 2D graph (6 expressions) | 12 s, shared | 15 s | 82 s | 204 s |
| 3D graph (4 expressions) | 15 s, shared | 18 s | 70 s | 180 s |
| Lists, definitions, `init` | — | — | 30 s | 120 s |

- The **watchdog** thread inside the engine checks every 0.2 s. When a deadline has passed, it writes a reply with `timeout:true` and `restart:true` and exits. This is faster than waiting for libqalculate's own abort.
- **Java** ( `Engine` ) runs one engine process and handles one request at a time. If no reply arrives in time, it kills the process and starts a new one for the next request.

| Situation | What Java does | What the page gets |
|---|---|---|
| The reply has `restart:true` | Passes the reply on, then stops the process | The engine's reply |
| No reply in time | Kills the process | Code `timeout` |
| The user cancels (taps "calculating…") | Kills the process | Code `cancelled` |
| The engine exits while working on a request | Never resends that request, because it would probably crash again | Code `crash` |
| Writing the request fails because the engine had already exited | Starts a new engine and sends the request once more | The normal reply, or code `engine` |

After a restart, Java restores `ans` with `set_ans`, and libqalculate reloads the user definitions from its files. The user notices nothing, except that `ans2` to `ans5` and temporary variables (such as `x = 5`) are gone.

Since 1.2.0, `set_ans` reads the value with as many digits of precision as it has: a new engine starts at precision 10, and before 1.2.0 a restored approximate `ans` kept only about 38 digits. Numbers of 2,000 digits or more are read directly with GMP and passed in as temporary variables, because libqalculate reads digits one at a time, in time that grows with the square of their number (200,000 digits: 0.2 s instead of 3.4 s on a PC, with exactly the same result).

## 7. Data, external interfaces and security

All data stays in the app's private storage (basic design 2.7). Every file is first written to a temporary file and then renamed, so a forced stop leaves either the old or the new content.

| Data | Where | Notes |
|---|---|---|
| History | `state/qalc.history.json` | Up to 300 items. The newest results are kept whole up to 1,000,000 characters in total; older long results keep their first 2,000 characters |
| Settings | `state/qalc.settings.json` | 20 items (lang and theme were added in 1.3.0, appearance in 1.4.0) |
| Display language | `state/qalc.lang.json` | "en" or "ja", written only when the user chooses with "Language / 言語"; English when missing (1.3.0) |
| Expression being typed | `state/qalc.draft.json` | Saved 0.5 s after typing |
| Graph inputs | `state/qalc.plot.*.json`, `state/qalc.plot3d.*.json` | Expressions, ranges, grid |
| `ans` | SharedPreferences `engine`, key `ans` | Up to 500,000 characters (an approximate value is saved as interval(lower, upper): about 400,000 characters for 200,000 digits) |

| Data | Where | Notes |
|---|---|---|
| QCalc Pro | SharedPreferences `pro` | owned, price, pre_pro, era, legacy, test_locked (1.3.0). The purchase itself is kept by Google Play on the user's Google account |
| User-defined variables and functions | `qalculate/definitions/*.xml` | libqalculate's XML format. Unreadable files are moved to `qalculate/broken-definitions/` |
| Exchange rates | `qalculate/eurofxref-daily.xml`, `btc.json`, `rates.json` | As downloaded |
| Exported history | Where the user chooses (Android's file picker) | CSV, UTF-8 with a byte order mark (1.3.0). Outside the app's storage, so it stays when the app is removed |

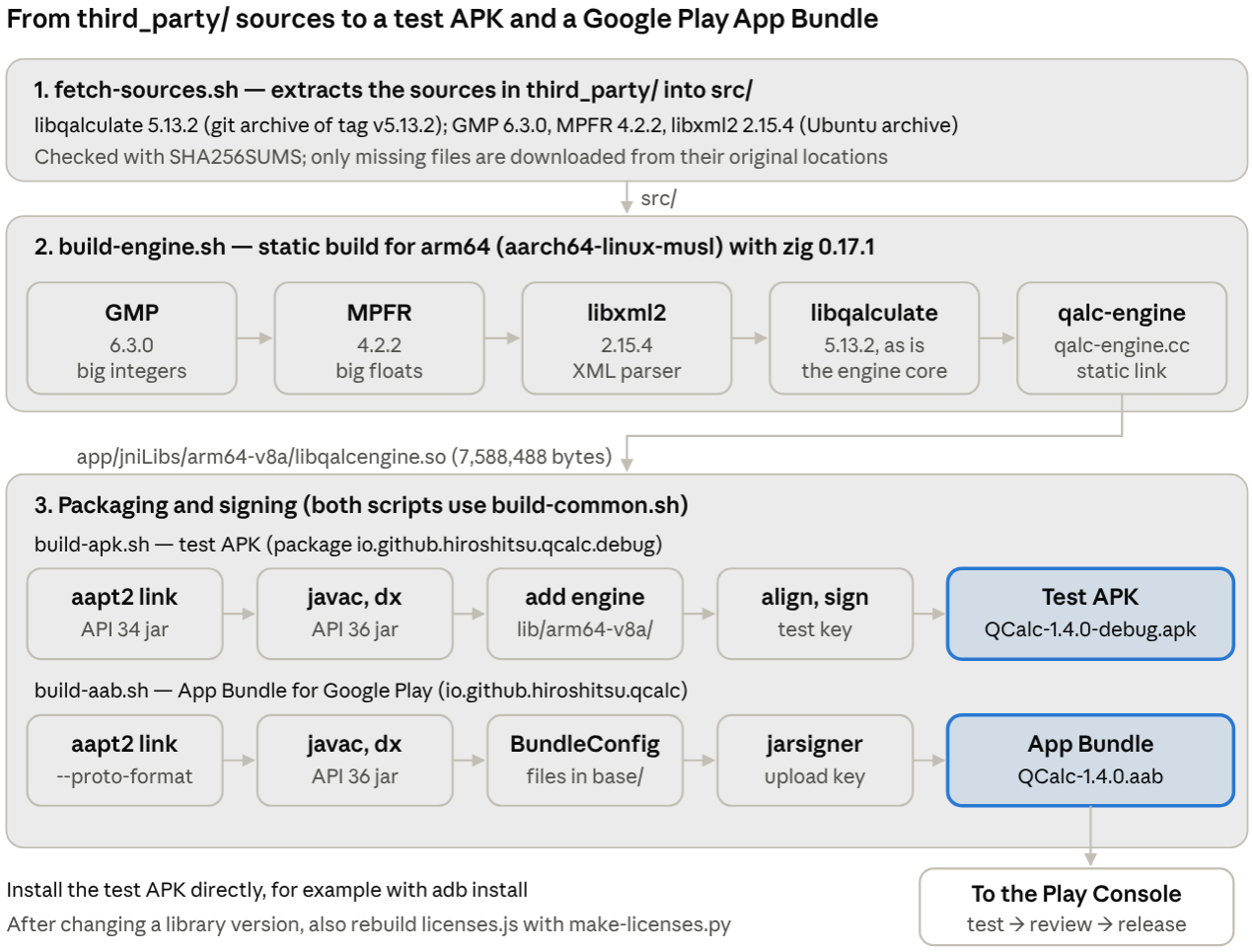
The app's only network traffic is downloading exchange rates over HTTPS from the same three sources that Qalculate! uses (basic design 2.8): the European Central Bank (major currencies), Coinbase (Bitcoin), and currency-api through jsDelivr (other currencies, with a fallback host). The User-Agent is `QCalc/1.4.0 (Android; libqalculate 5.13.2)`. Connections time out after 15 s and reads after 20 s, and replies over 1,000,000 bytes are discarded. In builds with billing, buying and checking QCalc Pro go through the Google Play Store app on the phone (detailed design 3.20).

Security and privacy (basic design 2.10):

- The WebView shows only the bundled pages under `file:///android_asset/`. File and content access are off, and other links open in the browser only if they are http or https.
- Results and user input are inserted with `textContent`, never as HTML.
- The JavaScript bridge checks every request as JSON and rejects requests that contain line breaks. Saved-state names are limited to 64 lowercase letters, digits and `._-`.
- A file that the page asks Java to save must be a CSV file with a plain name, and it is written only where the user picks in Android's file picker (detailed design 3.23).
- libqalculate is built with `--disable-insecure`, which disables `command()`. The engine reads and writes only `files/qalculate/`.

- The app has only two permissions (INTERNET and VIBRATE; builds with billing also get com.android.vending.BILLING from the Billing Library), collects no personal data, and has no ads or analytics. The privacy policy is at https://hiroshitsu.github.io/qcalc/privacy.html.
- The upload key and its password are never published or committed. Google keeps the app signing key.

# 8. Build and release




Build flow · three stages to the test APK and the App Bundle

Three shell scripts build everything on Ubuntu 24.04 (build and maintenance manual 5.2–5.6). Neither the Android NDK nor Gradle is needed. An Android Studio (Gradle) setup for the same sources is also included; it is also how the build with billing for QCalc Pro is made (qcalc.billing=true, build and maintenance manual 5.11). It has not been tested yet, because the build environment cannot reach Google's Maven repository.

- **Engine**: zig 0.17.1 cross-compiles GMP, MPFR, libxml2 and libqalculate as static libraries (-O2) for `aarch64-linux-musl`, then links `qalc-engine.cc` (C++17) statically with `-Wl,-z,max-page-size=16384`, so that the engine also runs on devices

with 16 KB memory pages. The result is a 7,588,488-byte executable.

- **libqalculate configure options**: `--enable-compiled-definitions` (definitions built into the library), `--disable-insecure` (no `command()`), `--without-libcurl` (Java downloads the rates), `--without-icu`, `--without-gnuplot-call` (the page draws the graphs), `--with-readline=no` and `--disable-nls`.
- **Outputs**: `QCalc-1.4.0-debug.apk` (3,319,892 bytes; a separate package, `io.github.hiroshitsu.qcalc.debug`, signed with a test key) and `QCalc-1.4.0.aab` (3,310,192 bytes, signed with the upload key, SHA-256 b7731e62bb83d30899eb76f8f714c437dab114836ae1bba8b9bd8a05026204e5), both without billing. The bundle is validated with bundletool 1.17.2. Its BundleConfig turns off "uncompressed native libraries", so that the APKs that Google Play makes still extract the engine as a file.
- **Updating libqalculate** (build and maintenance manual 5.9): put the new tagged source in `third_party/`, rebuild, port any upstream changes to the steps copied from `qalc` (`calculate_once`, `store_ans`), check the exchange-rate sources, regenerate the license texts, and rerun all the tests, including the comparison with `qalc`.
- **Releasing an update**: raise `VERSION_CODE` in `version.properties` (6 for 1.4.0), build the bundle, test the debug APK on a phone, and upload the bundle to a Play Console track.

## 9. Testing

No physical device was available, so each of the four parts was tested on a PC running Ubuntu 24.04 (test specification 6.1):

- **Engine**: the same arm64 engine binary that ships in the APK, run with `qemu-aarch64-static` and compared with the official `qalc` built from the same libqalculate tag.
- **Java**: `Engine.java` and `StateStore.java` run unchanged on OpenJDK 21 with small stand-ins for Android's `Context` and `SharedPreferences`, against both the real engine and a fake one.
- **Page**: the real `app/assets` in Chromium (Playwright) at the size of a phone screen, with requests relayed to the real engine.
- **Packages**: apksigner, zipalign, aapt2, jarsigner and bundletool on the APK and the App Bundle.

| Area | Items | Passed |
|---|---|---|
| Comparison with the official `qalc` | 317 | 317 |

| Area | Items | Passed |
|---|---|---|
| Engine functions, 3D graph values, calculus and matrices, Japanese units | 66 | 66 |
| All built-in examples (on the English and the Japanese screen) | 73 | 73 |
| Java (real engine and fake engine) | 24 | 24 |
| Page: keypads, calculus and matrices, examples, undo, text size, search, history, 3D graphs, fixes | 89 | 89 |
| APK and App Bundle checks | 22 | 22 |
| High precision (1.2.0): 200,000 digits compared digit by digit with mpmath, limits, restoring a long ans | 16 | 16 |
| Java (1.2.0): ans kept with all its digits | 7 | 7 |
| Page (1.2.0): long results shown, copied and saved | 32 | 32 |
| Page (1.3.0): QCalc Pro, color themes, CSV export, the preview, digit grouping | 19 | 19 |
| Page (1.3.0): the language of the screen, and no Japanese left on the English screen | 8 | 8 |
| Page (1.4.0): suggestions, the history, Light or dark, landscape, wrapping of long numbers | 27 | 27 |
| **Total** | **700** | **700** |

In addition, a build of the engine with AddressSanitizer and UndefinedBehaviorSanitizer ran the functional tests and 33,000 random requests, and the release engine ran 6,000 random requests, with no errors. For 1.3.0, 27 page tests were added (Q01 to Q27), and the comparison with qalc, the engine tests, the examples of both screens, the 31 Java tests, the page tests on the Japanese screen and the 22 package checks were run again with the 1.3.0 build (2026-11-13). For 1.4.0, 27 page tests (T01 to T27) and 8 expressions with “to mixed” were added, and all the tests except the high-precision ones were run again with the 1.4.0 build (2026-11-13), including the long-result and history page tests; a test with random taps on the page (350 actions with each of 3 seeds) found no script errors. The high-precision tests are results of 1.2.0 (test specification 6.1).

Not tested yet (test specification 6.6): installing and running on a real device, TalkBack, a real exchange-rate download, input with a real Japanese keyboard, rendering in the Android System WebView, window insets and predictive back on a device with Android 15

or later, the Android Studio build, a device with 16 KB pages, Google Play's review and closed test, buying QCalc Pro in a build with billing (with a license tester's account), the language and the CSV export on a phone, and on a phone the suggestions with the phone keyboard and the colors of the system bars with the Light or dark setting.

## 10. Components and versions

All calculation is done by the official libqalculate 5.13.2, whose source is not changed. The app's own code is the bridge program to the engine, the Java code for Android and the page. At every start, the bridge program adds the ten Japanese units and `trace()` to the running engine (libqalculate's files are not changed).

| Component | Version and settings | Source | License |
|---|---|---|---|
| App | QCalc 1.4.0 ( `io.github.hiroshitsu.qcalc` , versionCode 6), arm64-v8a. App Bundle for Google Play: 3,310,192 bytes | This project (published as HiroshiTsu/qcalc on GitHub) | GPL v2 or later |
| Calculation engine | libqalculate 5.13.2, with the definitions built in ( `--enable-compiled-definitions` ) and `command()` and the like disabled ( `--disable-insecure` ); no libcurl, ICU, gnuplot, readline or NLS | github.com/Qalculate/libqalculate (tag v5.13.2) | GPL v2 or later |
| Big integers | GMP 6.3.0 | The upstream tarball as Ubuntu distributes it (gmp_6.3.0+dfsg) | LGPL v3 or later |
| Big floats | MPFR 4.2.2 | The upstream tarball as Ubuntu distributes it | LGPL v3 or later |
| XML parser | libxml2 2.15.4 | The upstream tarball as Ubuntu distributes it | MIT |
| Cross compiler | zig 0.17.1 (pip's ziglang), target aarch64-linux-musl, static linking, aligned for 16 KB pages | PyPI | MIT |
| Android target | minSdk 24 (Android 7.0), targetSdk 36 (Android 16). Java is compiled against android.jar of API 36; aapt2 uses that of API 34 | Android SDK Platform 36 and 34 (not in the repository) | — |
| Signing | Play App Signing (Google keeps the app signing key). The upload key is RSA 4096 bits, certificate SHA-256 73:16:7B:03:1C:34:CC:0D:09:D2:A9:E1:D7:D1:E8:41:34:22:03:20:48:2A:B8:4B:80:0D:DB:0F:95:D2:E3:38. The test APK is signed with a separate test key | The project's upload key (kept in signing/, never published) | — |
| Android Studio settings | Android Gradle Plugin 8.11.1, Gradle 8.13, compileSdk 36 (building in Android Studio has not been tried) | Google Maven, Gradle | Apache 2.0 |
| C/C++ runtime libraries | musl libc, LLVM libc++, libc++abi, libunwind, compiler_rt (bundled with zig 0.17.1, statically linked) | zig's distribution | MIT, Apache 2.0 with LLVM Exceptions |
| Billing (builds with billing only) | Google Play Billing Library 8.3.0 (included only when Gradle runs with qcalc.billing=true) | Google Maven | Google's terms (not open source; QCalc's own code carries an exception that allows linking it) |

# 11. Guide to the documents

| Document | English | Japanese | What it covers |
|---|---|---|---|
| Overview | Design overview (English) (this tab), 00-overview-en.pdf | 概要, 00-overview.pdf | What QCalc is and what changed in 1.4.0, 1.3.0, 1.2.0 and 1.1.0; the design in brief; components and versions; glossary; revision history |
| 1. Requirements definition | 1. Requirements, 01-requirements-en.pdf | 1. 要件定義書, 01-requirements.pdf | Purpose, scope, 22 use cases, 35 functional requirements, non-functional requirements, constraints, 22 acceptance criteria, QCalc Pro |
| 2. Basic design | 2. Basic design, 02-basic-design-en.pdf | 2. 基本設計書, 02-basic-design.pdf | Components, deployment, modules, screens and transitions, keypads, main flows, data, external interfaces, error handling, security |

| Document | English | Japanese | What it covers |
| --- | --- | --- | --- |
| 3. Detailed design | `3. Detailed design`, 03-detailed-design-en.pdf | `3. 詳細設計書`, 03-detailed-design.pdf | Class diagram, JavaScript bridge, engine protocol, calculation, engine supervision, page modules, input, history, graphs, exchange rates, data formats, time limits, error codes and messages, Android 13–16, licenses, App Bundle, search, additions to the engine, QCalc Pro, language, color themes, CSV export, suggestions, Light or dark |

| Document | English | Japanese | What it covers |
|---|---|---|---|
| 4. User manual | `4. User manual`, 04-user-manual-en.pdf | `4. 操作マニュアル`, 04-user-manual.pdf | Installation and the language, the screen, calculating (also very long results), calculus, matrices, keyboard input, examples, the menu (graphs, lists, QCalc Pro, exporting the history), settings, troubleshooting, with screenshots |
| 5. Build and maintenance manual | `5. Build and maintenance`, 05-build-and-maintenance-en.pdf | `5. ビルド・保守マニュアル`, 05-build-and-maintenance.pdf | Environment, build flow, sources, engine build, APK, App Bundle, releases, checks, updating libqalculate, maintenance notes, Android Studio and the build with billing |
| 6. Test specification and results | `6. Test specification and results`, 06-test-specification-en.pdf | `6. テスト仕様書・結果`, 06-test-specification.pdf | Test policy and environment, engine tests, Java tests, page tests, package checks (700 items), tests not yet done, known limits |

The PDF files are in the design/ folder of the source repository. The steps for publishing QCalc on Google Play (the entries in Play Console, testing and review) are kept in a separate working document that is not part of this set.

## 12. Glossary

| Term | Meaning |
|---|---|
| libqalculate | The calculation library of the official Qalculate!. It does every calculation in QCalc |
| Engine | The bridge program `qalc-engine` linked with libqalculate. It ships as `lib/arm64-v8a/libqalcengine.so` and runs as a child process |
| Preview | The result shown while an expression is being typed |
| Confirm | Pressing “=”: the expression is calculated with `store:true` and the result is added to the history |
| `ans` event | The line that the engine sends after a confirmed calculation, with the new `ans` ( `value` ) and the text to insert ( `input` ) |
| Watchdog | The thread inside the engine that answers with a timeout and exits when a calculation runs past its limit + 3 s |
| Message stop | libqalculate's way of holding back warnings and errors during a calculation ( `beginTemporaryStopMessages` ) |
| Template | A function call with its arguments laid out, inserted by a key (for example `integrate(, 0, 1)` ), with the cursor where the expression goes |
| Upload key | The key that signs the App Bundle uploaded to Google Play. Google keeps the key that signs the app itself (Play App Signing) |
| Precision | The number of significant digits used to calculate and show results: 2 to 200,000 in the settings (10 by default). Raised from 1,000 to 200,000 in 1.2.0; since 1.3.0, up to 1,000 without QCalc Pro |

| Term | Meaning |
| --- | --- |
| QCalc Pro | A one-time product in Google Play ( `qcalc_pro` ) that unlocks precision above 1,000 digits, 3D graphs, color themes and exporting the history (1.3.0) |
| Earlier user (legacy) | Someone who used QCalc before Pro began (the phone has a saved history or settings, or ran a build without billing). They keep the Pro features for free, also after updating to a build with billing. Recognized by a mark kept on the phone, which clearing the app's data removes |
| Display language ( `qalc.lang` ) | The language of the screen. English is the default; the choice is saved in qalc.lang only when the user picks a language with “Language / 言語” (1.3.0) |
| Build with billing | A build made with Gradle and qcalc.billing=true. It contains PlayBilling and the Google Play Billing Library 8.3.0, so QCalc Pro can be bought. The shell scripts make builds without billing, which lock nothing |
| Suggestions | Items shown under the input field that are typed in when tapped. Two kinds: conversion targets after “to” (units, or ways to write a number), and names (functions, variables, units) while typing with the phone keyboard (1.4.0) |
| Light or dark (appearance) | The light and the dark colors of the screen. The setting offers Same as the phone (the default), Light and Dark (1.4.0). It is separate from the color theme, which sets the color of the keys and the highlights |

# 13. Revision history

| Date | What | Changes |
|---|---|---|
| 2026-11-13 | Design documents, 6th edition | Revised the overview and the six documents for QCalc 1.4.0, in English and in Japanese. Added the suggestions (detailed design 3.24), the Light or dark setting (3.25), the date and time and the update of the history, the fixes such as “to mixed”, and versionCode 6. The user manual explains the suggestions with a new screenshot, and the screenshots of the screens that changed (the settings, the menu of a history line, the graphs, a very long result) were taken again. The test specification gained the 27 tests of 1.4.0 and the 8 “to mixed” comparisons, and shows the results of running the tests again with 1.4.0 |
| 2026-11-13 | App QCalc 1.4.0 | A review of the whole app (versionCode 6). Suggestions under the input field (conversion targets after “to”, names while typing with the keyboard), the Light or dark setting, the date and time in the menu of a history line. Fixed “to mixed” giving results with Japanese units, a meaningless preview for an expression ending with “to”, and the keys being pushed off the screen by a long expression in landscape. The history is no longer built again at every calculation |
| 2026-11-13 | Design documents, 5th edition | Revised the overview and the six documents for QCalc 1.3.0, and made English versions of all the documents (seven English tabs, with the same content as the Japanese ones). Added the English and Japanese screens, QCalc Pro, color themes, exporting the history, all digits in the preview, digit grouping and versionCode 5. Redrew the diagrams and added English versions of them. Added the 27 tests of 1.3.0 to the test specification and updated it with the results of 1.3.0 |

| Date | What | Changes |
|---|---|---|
| 2026-11-13 | App QCalc 1.3.0 | English and Japanese screens (English by default; “Language / 言語” switches), QCalc Pro (a one-time purchase; the build with billing is made with Gradle), seven color themes, exporting the history (CSV). The preview while typing also shows all the digits (the preview of 4587! had shown only its beginning), and digit grouping puts commas only in the integer part. versionCode 5, together with the fix that makes English the default |
| 2026-11-12 | Design documents, 4th edition | Revised the overview, the six documents and the English overview for QCalc 1.2.0: the precision limit (200,000 digits), showing, copying and saving long results, carrying ans over with all its digits, and what happens when copying fails. Added “Very long results” and screenshots to the user manual and retook the screenshots of the settings. Added 55 tests of high precision, Java and the page to the test specification |
| 2026-11-12 | App QCalc 1.2.0 | Raised the precision limit from 1,000 to 200,000 digits (versionCode 3). Long results are shown and copied with all their digits, and the history keeps the newest ones up to 1,000,000 characters in total. Fixed ans being rounded to about 38 digits after the app closed, and made restoring a long ans fast. The app no longer stops when a result is too long to copy |
| 2026-11-12 | Design documents, 3rd edition | Revised the overview and the six documents for QCalc 1.1.0: the key tabs (calculus and matrices), matrices as tables, examples, undo, text sizes, search in Japanese, the Japanese units and trace(), and the workaround for the hidden warnings. Redrew the diagrams (the keys, the screen layout and others), added diagrams such as “Changes in 1.1.0”, and retook the 14 screenshots (27 screens) of the user manual |

| Date | What | Changes |
|---|---|---|
| 2026-11-12 | App QCalc 1.1.0 | Easier to use (versionCode 2): four key pages (123, Functions, Calculus, Matrix), so that derivatives, integrals, limits, differential equations and matrices can be typed with keys; matrix results as tables; the a=b key, examples and tips, undo, text sizes, Japanese names and search, the Japanese units, trace(). Fixed warnings that disappeared after a definite integral with an infinite bound, the text inserted from the unit list, when exact forms are shown, old coordinates in graphs, stopping a calculation and more |
| 2026-11-12 | Design documents, 2nd edition | Revised the overview and the six documents for QCalc 1.0.0: 3D graphs, Android 15 and 16, the license screen, the App Bundle and signing, the Google Play materials, review fixes and test results, with new diagrams and screenshots |
| 2026-11-12 | App QCalc 1.0.0 | The version for Google Play (2026-11-11 to 12): the name QCalc and the package name io.github.hiroshitsu.qcalc, a new icon, targetSdk 36 (screen edges and the back gesture), 3D graphs, the open source license screen, a link to the privacy policy, a fix for the newest history line being hidden, and the App Bundle (Play App Signing). The last review fixed details of the cursor keys, the graph messages and 3D graphs, and a test APK that did not start (the build named the wrong activity to start) |
| 2026-11-09 | Source set | Added settings to open the sources in Android Studio (Gradle). Put the resources and the built engine under app/ and the version in version.properties. Checked that the contents of the APK did not change (the first edition's tests) |
| 2026-11-09 | Design documents, 1st edition | Wrote the six documents, from the requirements to the tests, for the source code of app 5.13.2-3, with 13 diagrams (use cases, components, deployment, classes, sequences, states, screen transitions and others) and 7 screenshots (13 screens) in the user manual |

| Date | What | Changes |
|---|---|---|
| 2026-11-09 | App 5.13.2-3 | Fixes from three independent code reviews (Android, page, engine): assignments are not run in the preview, one evaluation so that ans equals what is shown, detecting timeouts, supervising and restarting the engine, exact reuse of the history, converting Japanese input, the local currency (yen), saving the state at once and more |
| 2026-11-09 | App 5.13.2-2 | The version handed out before (signed with the same key, so it can be updated in place) |